Gray-Zone Maritime Control in the First Island Chain: A Networked Escalation Game

Updates to the Workshop 1 Writeup (working copy for Workshop 2)

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This is a copy of the Workshop 1 report (as submitted, branch `workshop1-final`) with the Workshop 2 model changes written in. Text in this color is new or changed. ~~Struck-out red text~~ is removed. Sections not yet touched are unchanged from Workshop 1, including all results, which will be replaced after the rerun.

1 Setting and research questions

Contested system: China and the United States compete, *below the threshold of war*, for day-to-day control of the waters inside the First Island Chain. China claims most of the South China Sea and contests the Senkakus, pressing these claims through the persistent presence of coast guard and maritime militia (CCG) hulls backed by navy (PLAN) ships. The U.S. rejects maritime claims beyond what international law allows and keeps the seas open with a smaller number of destroyers (DDG) on presence and freedom-of-navigation patrols (Helmus et al., 2024; Green et al., 2017). Every two days, each side decides *where* to station its ships on a network of zones and *how* to act there: patrol (P) or assert (A: shadowing, close maneuvering, querying and warning, and, for the U.S. only, freedom-of-navigation operations).

Why strategic interaction is central: control of a zone depends on the ratio of local forces, so each side's best allocation depends on the other's. Escalation risk is shared. Incidents need both sides present, and war costs both. Neither side sees the other's entire fleet disposition, only forces present where they also maintain forces.

Research questions:

RQ1. *Control vs. war risk.* Each side caps how much incident risk it will accept each period (its risk budget ε_i). How much more control does a side gain by accepting more risk, how much more likely does war become, and what risk level does each side pick when it knows the other side is picking too?

RQ2. *Game theory vs. rules of thumb.* Does a side that plans with game theory (best responses, equilibrium or maximin plans) do better than one that follows simple rules or moves at random? Can both sides keep patrolling over many periods because asserting once would end the truce and push both into riskier play from then on?

RQ3. *Weather and redesign.* Does the plan or Blue redesign that works best in normal weather still work best across calm, typhoon and extreme weather scenarios, both on average and in the worst scenario?

Outcomes (averaged over 200 simulated games per setup): Blue's value-weighted control share $\bar{c}_{\mathsf{B}} = \frac{1}{\tau}\sum_t \sum_v V_v^{\mathsf{B}} c_v^t / \sum_v V_v^{\mathsf{B}} \in [-1,1]$, war rate and payoffs $u_{\mathsf{B}}, u_{\mathsf{R}}$. For RQ3, also expected and worst-case payoff over ω.

Changes from the proposal and how we used the feedback: the model is now a formal game on a graph (§3). The best response is an exact MIP whose feasible region is a risk budget, with the proposal's commander types as budget levels (§3.7). The RQs now ask for the budget-game equilibrium, equilibrium play, and robustness and redesign. We define three Blue redesigns (§3.10), propose values for missing data (Tables 1–3), report analytical results and 77 configurations × 200 seeds (§4), and add weather scenarios scored by expected and worst-case payoff, as Prof. Sapsis asked (§3.2, §4.4).

2 Literature and current practice

Documented practice: Helmus et al. (2024) describe China's use of coast guard and maritime militia vessels to maintain presence and obstruct maritime activities while seeking to avoid a conventional military response. Green et al. (2017) emphasize credible commitments, coordinated allied responses, and acceptance of calculated escalation risk while exercising restraint. U.S. practice is routine presence and freedom-of-navigation patrols, with encounters managed by the Code for Unplanned Encounters at Sea (WPNS, 2014). From these sources we take the actions, the asymmetric hull types, and the idea that incidents, not shots, drive escalation.

Models: Wasser et al. (2019) developed a structured gray-zone game with explicit objectives and probabilistic adjudication. We automate the rules to support repeated simulation and strategy comparison. Shieh et al. (2012) document a deployed Stackelberg model of randomized Coast Guard port patrols. Washburn and Wood (1995) model network interdiction as a two-person zero-sum game. Schelling (1966) represents escalation as a shared, manipulated risk, motivating our incident hazard.

What we add: we combine maritime allocation, evolving control, and shared escalation risk in a general-sum game where both sides contest control simultaneously and war costs both. Computational rules support comparisons of strategies and designs across thousands of runs. Numerical parameters are our assumptions (Table 1), not values derived from documented literature.

3 Game formulation

One period t (timeline): (1) Nature draws the sea state w_r^t of each region (scenario ω). (2) Each player observes its own ships, ℓ^t, c^t, w^t, and detects opponent ships w.p. $q_{i,v}$. (3) Ships whose endurance is nearly spent are forced toward homeor toward their side's oiler. Ships due for maintenance or with an engineering casualty are forced home. (4) Both players choose moves and postures

simultaneously. (5) Moves resolve under movement budgets, and endurance updates. Maintenance clocks, engineering casualties and repairs update, and each oiler moves. (6) Control indices update, incidents are drawn, and ℓ moves up or down. (7) Stage payoffs are paid, and the game stops if $\ell = L$ (war) or $t = T = 60180$.

Symbol	Meaning	Value
T; δ	periods (2 days); discount	60180; 0.98
N_k	fleet size	10 DDG; 8 PLAN, 6 CCGTable 2
D_k	endurance (periods)	10 DDG; 8 PLAN; 6 CCGTable 2
F_i	movement budget (moves/period)	8 Blue; 10 Red23 Blue; 40 Red
w_k	strength per hull (rough seas)	1 DDG, PLAN; 0.6 CCG (0.3)Table 2
α	assert multiplier	1.5
κ; λ	presence damping; control rate	1; 0.3
h_0	baseline incident hazard	0.006
h_A^B; h_A^R	assert hazard	0.03; 0.036/0.012 PLAN/CCG0.03/0.01; 0.036/0.012 navy/coast guard
s_h	hazard scale	0.12

Symbol	Meaning	Value
L; ℓ^1	war level; starting level	5; 1
μ	chance quiet period lowers ℓ	0.25 (D3: 0.5 at $\ell \geq 3$)
γ; η	move; tension cost	0.1; 1
W	war penalty (both sides)	200
$q_{i,v}$	detection: here / nearby / far	0.95 / 0.6 / 0.2, $\times$0.7 rough
Λ; β	MIP: incident price; persistence	20; 0.96
ε_i	escalation-risk budget	0.05 / 0.15 / 0.40
ζ_D	redesign cost per period	D1 0.3; D2 0.5; D3 0
M_{on}; M_{in}	periods operating; in maintenance	90; 45
m_0	share in maintenance at $t = 1$	1/3
p_c; Z_c	casualty chance per period at sea; repair periods	$1 - 0.99^2 \approx 0.02$; 5
D_o	oiler endurance (periods)	15

Table 1: Parameter values, all of them our assumptions (no public data). Hazards and μ were set so that war is uncommon but not rare (0–34% of games).

3.1 Players, map and time

Players: $\mathcal{I} = \{\mathsf{B}, \mathsf{R}\}$. Blue is the U.S. Navy, $\mathcal{K}_\mathsf{B} = \{\text{DDG}\}$. Red is China, with navy ships and coast guard or maritime militia hulls, $\mathcal{K}_\mathsf{R} = \{\text{PLAN}, \text{CCG}\}$ (fleets in Table 1). Blue is the United States with Japan and the Philippines, with eight ship types $\mathcal{K}_\mathsf{B}$. Red is China with North Korea, with six types $\mathcal{K}_\mathsf{R}$ (Table 2). Each type is navy or coast guard. Coast guard and maritime militia hulls are grouped as coast guard.

Side	Nation	Type k	Ships	w_k	D_k	Real / game	Home ports (real)
Blue	U.S.	US-DDG	destroyers and cruisers, 7th Fleet	1.0	10	12 / 4	YOK 12
		US-LCS	littoral combat ships*	0.5	5	2 / 1	GUA 2
	Japan	JP-DDG	JMSDF destroyers	1.0	10	26 / 9	YOK 13, SAS 13
		JP-FF	JMSDF frigates and escorts	0.7	8	16 / 5	SAS 16
		JCG	Japan Coast Guard large patrol vessels	0.6	6	12 / 4	ISG 12
	Philippines	PH-FF	Philippine Navy frigates	0.7	8	4 / 1	PHL 4
		PH-PS	corvettes and offshore patrol vessels*	0.4	5	9 / 3	PHL 9
		PCG	Philippine Coast Guard cutters*	0.6	6	6 / 2	PHL 6
Red	China	CN-DDG	PLAN destroyers	1.0	10	30 / 10	QDO 8, NBO 10, SAN 12
		CN-FF	PLAN frigates	0.7	8	27 / 9	QDO 7, NBO 9, SAN 11
		CN-FS	PLAN corvettes*	0.4	5	25 / 8	QDO 6, NBO 9, SAN 10
		CCG	China Coast Guard large cutters*	0.6	6	75 / 25	QDO 19, NBO 26, SAN 30
	North Korea	KP-DDG	destroyers	1.0	8	3 / 1	NAM 3
		KP-FS	light frigates and corvettes*	0.3	5	11 / 4	NAM 11
Totals						Blue 87 / 29, Red 171 / 57	

Table 2: Fleets in theater (no aircraft carriers). Counts are our estimates of the ships each nation keeps in or near the theater, from open sources (O'Rourke, 2025; Wikipedia, 2026a,b; AMTI, 2026; Wikipedia, 2026c; Janes, 2026; Wikipedia, 2026d). The Chinese side is about half of the PLAN and CCG totals, for the share facing this theater. w_k is the strength weight (destroyers and cruisers highest), D_k the endurance in two-day periods. Types marked * are small hulls whose strength is halved in rough seas. "Game" is the number of game ships, each standing for three real ships (§3.1).

Game ships stand for three real ships: the best-response program (§3.8) takes too long to solve at full size. We therefore play with one game ship for every $s = 3$ real ships of the same type. A game ship has weight $s\,w_k$, so the local strengths $S_{i,v}$, and with them control and incident hazards, stay on the scale of the real fleets. Each type's count is divided by s and rounded, with at least one ship per type, and split over home ports by largest remainder:

$$N_k^{\text{game}} = \max\big(1,\ \text{round}(N_k/s)\big), \qquad N_{k,h}^{\text{game}} = \lfloor N_{k,h}/s \rfloor + \mathbf{1}\big[h \text{ is among the } N_k^{\text{game}} - \textstyle\sum_{h'} \lfloor N_{k,h'}/s \rfloor \text{ ports with the largest remainders}\big].$$

Every ship-level rule (endurance, maintenance, engineering casualties, oiler) applies to a game ship as a whole. The movement budget counts game-ship moves, $F_i = \text{round}(f_i N_i^{\text{game}})$ with $f_\mathsf{B} = 0.8$ and $f_\mathsf{R} = 0.7$.

Map: the theater is an undirected graph $G = (\mathcal{V}, \mathcal{E})$. Its nodes are home ports, where ships refuel, and eight contested zones $\mathcal{C} = \{\text{SEN}, \text{MIY}, \text{TWN}, \text{EOT}, \text{LUZ}, \text{SCB}, \text{STS}, \text{PAR}\}$ (Senkakus, Miyako Strait, Taiwan Strait, east of Taiwan, Luzon Strait, Scarborough Shoal, Second Thomas Shoal, Paracels). Blue's home ports are $H_\mathsf{B} = \{\text{YOK}, \text{GUA}, \text{SAS}, \text{ISG}, \text{PHL}\}$ (Yokosuka, Guam, Sasebo, Ishigaki, Subic Bay) and Red's are $H_\mathsf{R} = \{\text{QDO}, \text{NBO}, \text{SAN}, \text{NAM}\}$ (Qingdao, Ningbo, Sanya, Nampo)[1]. Ports are national. A ship of nation a refuels and does maintenance only at its nation's ports H_a: U.S. ships at YOK and GUA, Japanese ships at YOK, SAS and ISG, Philippine ships at PHL, Chinese ships at QDO, NBO and SAN, North Korean ships at NAM. Design D1 also opens PHL and SAS to U.S. ships. The new ports join the zones next to them (PHL to SCB, LUZ, STS. SAS to SEN, MIY. ISG to SEN, MIY, EOT,

[1] NBO and SAN stand for the Ningbo–Zhoushan and Hainan (Yulin) basing areas. CCG and militia ports are folded into them.

TWN. QDO to SEN, NBO, NAM). An edge joins two nodes a ship can sail between in one period. The zones form three regions that share weather: North $= \{\text{SEN}, \text{MIY}\}$, Center $= \{\text{TWN}, \text{EOT}, \text{LUZ}\}$, South $= \{\text{SCB}, \text{STS}, \text{PAR}\}$. Zone v is worth V_v^i to player i (Table 3, e.g. Scarborough Shoal and the Paracels matter more to Red).

Time: periods $t = 1, \ldots, T$ of two days, $T =$ ~~60 (about four months)~~180 (one year), discounted by δ per period. All ships start in port with full endurance, ~~split evenly between each side's two ports~~at their home ports (Table 2). A third of each fleet starts in maintenance (§3.4).

Zone	TWN	SEN	MIY	EOT	LUZ	SCB	STS	PAR
V_v^{B}	10	6	4	4	6	5	5	2
V_v^{R}	10	6	3	4	5	7	6	6
c_v^1	0	0.2	0.3	0.2	0.1	−0.2	0	−0.4

ω	calm	nominal	typhoon	extreme
π_ω	0.25	0.40	0.25	0.10
$p_s(\omega)$	0	0.02	0.06	0.12
$p_r(\omega)$	0	0.20	0.35	0.50

Table 3: Our assumed values. Left: zone values and starting control. Right: scenario weight π_ω and per-period chances of a storm starting $p_s(\omega)$ and of rough seas $p_r(\omega)$.

3.2 State, information and weather

State: $s^t = (x^t, d^t, c^t, \ell^t, w^t, r^t, z^t, e^t, o^t, d^{o,t})$. Here $x_{k,v}^t \in \mathbb{Z}_{\geq 0}$ is the number of type-k ships at node v. The *endurance* $d_j^t \leq D_k$ is the number of periods ship j can still stay at sea (down one per period at sea, refilled after a full period in port). The *control index* $c_v^t \in [-1, 1]$ of each zone is $+1$ when Blue is fully in charge and -1 when Red is. The *escalation level* $\ell^t \in \{0, \ldots, L\}$ runs from calm to armed conflict. The sea state of region r is $w_r^t \in \{\text{calm}, \text{rough}, \text{storm}\}$. Each ship j also carries a readiness state: r_j^t is the number of operating periods left before it is due for maintenance, z_j^t is the number of periods it still has to spend in port (maintenance or repair, 0 when it is available), and $e_j^t \in \{0, 1\}$ is 1 if it has an engineering casualty and is heading home. Each player's oiler is at node o_i^t with $d_i^{o,t}$ periods of its own fuel left (§3.4).

Information: each player sees its own ships and the public c^t, ℓ^t, w^t, but not where the opponent's ships are now. Each period it detects the opponent at node v with probability $q_{i,v}$ (Table 1, for an own ship there, a ship next door or v near a home port, or satellite only), seeing the count of each type and the posture. Its *belief* $\hat{x}_{-i,v}^t$ is the last count it detected at v (exact at the start, when all ships are in port), so moves are chosen at information sets that do not reveal where the opponent is now.

Weather: a random factor neither side controls. In each of four scenarios (not calendar seasons) $\omega \in \Omega = \{\text{calm, nominal, typhoon, extreme}\}$ with weights π_ω, each period a region not already in a storm enters one with probability $p_s(\omega)$, and is otherwise rough with probability $p_r(\omega)$ (Table 3). A storm lasts three periods. It closes the region's zones (no entry, ships inside stay), stops incidents there and freezes their control. Rough seas halve ~~CCG~~small-hull strength and degrade detection.

3.3 Actions and constraints

Actions: each period both players choose, simultaneously, a destination for each ship (stay, or sail one edge) and a posture $p_{i,v} \in \{\text{P}, \text{A}\}$ in each zone v where they will have ships. *Patrol* is routine presence. *Assert* stands for the §1 actions, treated as one posture. Asserting multiplies the side's strength there by α but makes incidents more likely (§3.5).

Constraints: with $f_{k,uv}^t$ the number of type-k ships sailing $u \to v$ in period t and $\mathcal{N}(u)$ the neighbors of u,

$$x_{k,u}^{t+1} = x_{k,u}^t + \textstyle\sum_{v \in \mathcal{N}(u)} f_{k,vu}^t - \sum_{v \in \mathcal{N}(u)} f_{k,uv}^t \qquad \text{(flow conservation: no ships are added or removed)}$$

$$\textstyle\sum_{k \in \mathcal{K}_i} \sum_{(u,v)} f_{k,uv}^t \leq F_i \qquad \text{(movement budget: at most } F_i \text{ moves per period, Table 1)}$$

where the theater-wide movement budget F_i stands in for operational requirements and station-keeping. *Endurance:* a ship may not sail to a node from which it could not ~~get home~~reach a refuel point on its remaining endurance, unless the move brings it closer to ~~home~~one. The refuel points of player i are its home ports and its oiler when the oiler is in a zone, $\mathcal{R}_i^t = H_i \cup \{o_i^t : o_i^t \in \mathcal{C}\}$. A ship uses whichever is closer, at distance $\text{dist}(v, \mathcal{R}_i^t) = \min_{u \in \mathcal{R}_i^t} \text{dist}_i(v, u)$. A ship whose endurance has fallen to its distance from ~~home~~the nearest refuel point is on *forced return*. It sails one edge toward ~~port~~that point each period, then spends one full period in port or alongside the oiler to refit, after which d_j resets to D_k. Forced moves use the movement budget first, and the player decides only for its *free* ships. *Ports, storms and infeasible orders:* no ship may enter opponent home ports or storm regions, and orders over budget are dropped.

3.4 Readiness: maintenance, engineering casualties and oilers

A year-long game needs the things that keep a real fleet from being at sea all the time.

Maintenance: a ship operates $M_{\text{on}} = 90$ periods (six months) and then spends $M_{\text{in}} = 45$ periods (three months) in maintenance at a home port, so in steady state a third of each fleet is unavailable. While a ship is available its maintenance clock runs down, $r_j^{t+1} = r_j^t - 1$. When $r_j^t \leq 0$ the ship is on forced return to its nearest home port (not to the oiler). On arrival it enters maintenance, $z_j = M_{\text{in}}$, and z_j falls by one each period. When z_j reaches 0 the ship is available again with $r_j = M_{\text{on}}$ and full endurance $d_j = D_k$. A ship with $z_j > 0$ stays in port and is not part of either player's decision.

Staggered start: at $t = 1$ a share $m_0 = 1/3$ of each player's ships, chosen at random, is in maintenance with z_j^1 drawn uniformly from $\{1, \ldots, M_{\text{in}}\}$. Every other ship has r_j^1 drawn uniformly from $\{1, \ldots, M_{\text{on}}\}$, so ships come due at different times and not all at once. These draws use their own random stream, so a given seed has the same maintenance schedule for every pair of policies.

Engineering casualties: each day at sea a ship suffers an engineering casualty with probability 1%, so in a two-day period the chance is $p_c = 1 - 0.99^2 \approx 0.02$. Then $e_j = 1$ and the ship is on forced return to its nearest home port. On arrival it is repaired for $Z_c = 5$ periods (ten days), $z_j = Z_c$, after which $e_j = 0$ and $d_j = D_k$. A ship that has a casualty while heading in for maintenance is repaired during the maintenance period.

Oilers: each player has one oiler. It is not commanded ship by ship. It follows a fixed rule that keeps it where its side is doing well. The side's *best region* is the one with the highest value-weighted control from that side's point of view,

$$r_i^{*,t} = \arg\max_{r \in \{\mathrm{N,C,S}\}} \sigma_i \frac{\sum_{v \in r} V_v^i c_v^t}{\sum_{v \in r} V_v^i}.$$

The oiler picks a zone of $r_i^{*,t}$ uniformly at random as its target, sails one edge per period toward it, and picks a new random target when it arrives or when the best region changes. So it wanders among the zones of the best region and follows when that region changes. Its own fuel d_i^o falls by one per period away from port. When $d_i^{o,t} \le \mathrm{dist}_i(o_i^t, H_i)$ it heads for its nearest home port, and one period there refills it to $D_o = 15$ (30 days). A ship that spends a full period in the oiler's zone without moving is refueled to D_k, as in port. The oiler obeys the same storm and port rules as warships. It adds no strength and cannot be attacked yet.

3.5 How the state changes: control, incidents and escalation

After the moves, steps 1–3 apply in each contested zone v not in a storm. Step 4 updates ℓ once for the theater.

1. Local strength: each side's weight of presence in the zone is $S_{i,v} = \alpha^{\mathbf{1}[p_{i,v}=\mathrm{A}]} \sum_{k \in \mathcal{K}_i} w_k x_{k,v}$, its hulls weighted by w_k (a CCG hull counts less than a navy ship, less still in rough seas), times α when asserting.

2. Control: the *local balance* $\phi_v = (S_{\mathsf{B},v} - S_{\mathsf{R},v})/(S_{\mathsf{B},v} + S_{\mathsf{R},v} + \kappa)$ lies between -1 and 1 and is positive when Blue is locally stronger. The constant $\kappa = 1$ keeps a small presence from counting as full control (e.g. one DDG alone gives $\phi_v = 0.5$). Control is earned by staying present and fades when a side leaves. Each period it moves a fraction $\lambda = 0.3$ of the way toward the local balance,

$$c_v^{t+1} = (1 - \lambda)\, c_v^t + \lambda\, \phi_v. \tag{1}$$

3. Incidents: an incident (a collision, an unsafe interaction, negative media coverage) can happen only where both sides have ships. It occurs with probability $\rho_v = 1 - e^{-h_v}$, where the *hazard* h_v is

$$h_v = s_h \big(h_0 + h^{\mathsf{B}}_{\mathrm{A},v} \mathbf{1}[p_{\mathsf{B},v} = \mathrm{A}] + h^{\mathsf{R}}_{\mathrm{A},v} \mathbf{1}[p_{\mathsf{R},v} = \mathrm{A}]\big) \cdot \min\{S_{\mathsf{B},v}, S_{\mathsf{R},v}\} \cdot \big(1 + \ell^t/2\big). \tag{2}$$

(i) A small baseline hazard h_0, plus a term for each side that asserts. ~~Red's assert hazard $h^{\mathsf{R}}_{\mathrm{A},v} = \theta_v h^{\mathsf{R}}_{\mathrm{A,PLAN}} + (1-\theta_v) h^{\mathsf{R}}_{\mathrm{A,CCG}}$ blends its navy and coast guard values by θ_v, the PLAN share of Red's strength at v~~Each side's assert hazard $h^i_{\mathrm{A},v} = \theta_{i,v} h^i_{\mathrm{A,navy}} + (1-\theta_{i,v}) h^i_{\mathrm{A,cg}}$ blends its navy and coast guard values by $\theta_{i,v}$, the navy share of its strength at v, so coast guard pressure is less escalatory. The scale s_h multiplies all hazards. It is set so that rule-based play on both sides ends in war in about 5% of one-year games in nominal weather. (ii) More ships in contact mean more encounters, limited by the smaller force. (iii) Incidents are likelier when tension is high (2.5× at $\ell = 3$).

4. Escalation: each incident raises ℓ by one. A period with no incident anywhere lowers ℓ by one with probability μ (Table 1). If ℓ reaches L, the crisis has become an armed conflict and the game ends.

3.6 Payoffs

Stage payoff: at the end of period t, player i receives

$$g_i^t = \sigma_i \sum_{v \in \mathcal{C}} V_v^i c_v^{t+1} - \gamma m_i^t - \eta\, \ell^{t+1} - \zeta_D \mathbf{1}[i = \mathsf{B}]. \tag{3}$$

Control: $\sigma_{\mathsf{B}} = +1$ and $\sigma_{\mathsf{R}} = -1$ turn the shared control index into each side's gain (e.g. Blue's control term is $V_v^{\mathsf{B}} c_v$, positive where Blue holds the zone and negative where Red does). *Fuel:* m_i^t ship moves in period t, at γ each. *Tension:* η per escalation level per period, for diplomatic and economic strain. *Redesign cost:* ζ_D is Blue's redesign cost per period (§3.10).

Total payoff: the game ends at τ, the first period with $\ell^\tau = L$ (war), or at T. Player i's payoff is the discounted sum of its stage payoffs, minus a war penalty W (Table 1) if the game ends in war:

$$u_i = \sum_{t=1}^{\tau} \delta^{t-1} g_i^t - W\, \mathbf{1}[\ell^\tau = L]. \tag{4}$$

Both sides pay W, so the game is general-sum, not zero-sum. Both prefer to avoid war. For scale, Blue earns at most $\sum_v V_v^{\mathsf{B}} = 42$ in a period in which it fully controls every zone, so a war ($W = 200$) costs roughly five such periods.

3.7 How a player decides: the best-response MIP with a risk budget

Our main decision rule, *MIP-BR*, is a best response to the opponent's believed positions that caps incident risk, written as a mixed-integer program and solved with Gurobi each period. It takes as given its beliefs $\hat{x}_{-i}$, the sea state, ℓ^t, its own ships, and, in each zone, the opponent posture it saw on its last detection there (patrol if never detected). Forced-return ships are fixed ($\bar{x}_{k,v}$, by destination).

Decision variables: $y_{j,w} \in \{0,1\}$ is 1 if free ship j goes to node w (only moves allowed by §3.3). Control and incident risk are nonlinear in ship counts, so we list every count a zone could end up with. Option $b_{v,(n,p)} \in \{0,1\}$ is 1 if the player will have $n = (n_k)_{k \in \mathcal{K}_i}$ hulls in zone v with posture p. Each option's value and risk are computed in advance, which keeps the program linear.

Value of an option:

$$a_v(n,p) = V_v^i \, \beta \, \sigma_i \phi_v(n,p) - \Lambda \rho_v(n,p), \qquad \beta = \frac{\lambda}{1-\delta(1-\lambda)} \approx 0.96.$$

with ϕ_v, ρ_v (§3.5) at believed opponent positions. Here β is the discounted effect on control of one period of balance, and Λ is the incident price.

The program:

$$\begin{aligned}
\max_{y,b} \quad & \textstyle\sum_{v \in \mathcal{C}} \sum_{(n,p)} a_v(n,p)\, b_{v,(n,p)} - \gamma \sum_j \sum_{w \neq \mathrm{loc}(j)} y_{j,w} - 1000\,\xi && \text{(control value} - \text{incident cost} - \text{fuel cost)} \\
\text{s.t.} \quad & \textstyle\sum_w y_{j,w} = 1 \quad \forall j && \text{(each free ship gets one destination)} \\
& \textstyle\sum_j \sum_{w \neq \mathrm{loc}(j)} y_{j,w} \le F_i - \text{(forced moves)} && \text{(movement budget)} \\
& \textstyle\sum_{(n,p)} b_{v,(n,p)} = 1 \quad \forall v && \text{(one option per zone)} \\
& \textstyle\sum_{(n,p)} n_k \, b_{v,(n,p)} = \bar{x}_{k,v} + \sum_{j \text{ of type } k} y_{j,v} \quad \forall v,k && \text{(the option matches the moves)} \\
& \textstyle\sum_{v \in \mathcal{C}} \sum_{(n,p)} \rho_v(n,p)\, b_{v,(n,p)} \le \varepsilon_i \,(1 - \ell^t/L) + \xi && \text{(escalation-risk budget)} \\
& y,\, b \text{ binary}, \ \xi \ge 0.
\end{aligned}$$

The escalation-risk budget: the left side of the last constraint is the expected number of incidents this period. The cap ε_i shrinks to zero as ℓ approaches war and defines the feasible region, the allocations the player considers safe enough. The proposal's conservative, moderate and aggressive commanders become three budgets (Table 1), and RQ1 varies ε_i. A ship on forced return may have to pass through a zone the opponent occupies. That risk cannot be avoided, and near war it can exceed the cap. The slack $\xi \ge 0$, penalized at 1000 per unit, lets the program exceed the cap by only that amount, so it always has a solution.

3.8 The best-response MIP for the Workshop 2 fleets

The program above is exact, but it does not scale to the Workshop 2 fleets. This subsection shows why and gives the version we use. The rules of the game (§3.3–§3.6) do not change. Only the way the player's decision problem is written down changes, and it stays exact.

Why the Workshop 1 program breaks: it lists every count vector a zone could end up with. If $A_{k,v}$ ships of type k could arrive at zone v this period, the number of options in that zone is

$$|\{b_{v,(n,p)}\}| \;=\; 2 \prod_{k \in \mathcal{K}_i} \big(A_{k,v} + 1\big).$$

In Workshop 1, $|\mathcal{K}_\mathrm{B}| = 1$ and $|\mathcal{K}_\mathrm{R}| = 2$, so this was a few hundred. With the eight Blue types at full size and, for example, $A_{k,v} = (12, 13, 16, 12, 2, 4, 9, 6)$, it is $2 \cdot 13 \cdot 14 \cdot 17 \cdot 13 \cdot 3 \cdot 5 \cdot 10 \cdot 7 \approx 8 \times 10^7$ options in one zone. The model cannot be built, whatever the license or the run time.

Step 1, moves as flows: free ships of the same type, at the same node, with the same set of allowed destinations are interchangeable. Call such a set a group g, with type $k(g)$, location $u(g)$, allowed destinations $\mathcal{W}_g$ and size $|g|$. Replace the per-ship binaries $y_{j,w}$ by integer flows

$$f_{g,w} \in \mathbb{Z}_{\ge 0} \ \ (w \in \mathcal{W}_g), \qquad \textstyle\sum_{w \in \mathcal{W}_g} f_{g,w} = |g|.$$

These are the flows $f^t_{k,uv}$ of §3.3. Every assignment of ships gives a flow and every flow gives an assignment, so this step changes nothing.

Step 2, options by navy and coast guard strength level: a zone's value depends on the player's ships there only through two numbers, its navy strength and its coast guard strength. Control uses their sum. The assert hazard also uses the navy share, because it blends the navy and coast guard values, $h^i_{\mathrm{A},v} = \theta_{i,v} h^i_{\mathrm{A,navy}} + (1-\theta_{i,v}) h^i_{\mathrm{A,cg}}$ (§3.5). Every weight $s\,w_k$, halved or not in rough seas, is a whole multiple of the unit $u = 0.05\,s$ ($u = 0.15$ for $s = 3$), so $u_k(w_r) = s\,w_k(w_r)/u$ is an integer. The navy and coast guard levels in zone v are

$$n_v = \sum_{k \text{ navy}} u_k \Big(\bar{x}_{k,v} + \sum_{g : k(g) = k} f_{g,v} \Big), \qquad c_v = \sum_{k \text{ coast guard}} u_k \Big(\bar{x}_{k,v} + \sum_{g : k(g) = k} f_{g,v} \Big),$$

so the own un-asserted strength is $R_v = u\,(n_v + c_v)$ and the navy share is $\theta_{i,v} = n_v/(n_v + c_v)$. The option $b_{v,(n,c,p)} \in \{0,1\}$ means "navy level n, coast guard level c and posture p in zone v". Only levels that the fixed ships and the arriving groups can actually produce are listed. The option values are the same formulas as before,

$$a_v(n,c,p) = V_v^i \, \beta \, \sigma_i \phi_v(n,c,p) - \Lambda \rho_v(n,c,p), \qquad S_{i,v} = \alpha^{\mathbf{1}[p=\mathrm{A}]}\, u\,(n+c),$$

with the hazard in ρ_v using $\theta_{i,v} = n/(n+c)$. Both control and incident risk are therefore exactly those of the Workshop 1 program. A patrol option does not depend on the split between n and c, because the navy share enters only the player's own assert hazard. So patrol options need only the total level, which keeps the program small.

The program:

$$\begin{aligned}
\max_{f,b}\ & \textstyle\sum_{v\in\mathcal{C}}\sum_{(n,c,p)} a_v(n,c,p)\, b_{v,(n,c,p)} - \gamma\sum_g\sum_{w\neq u(g)} f_{g,w} - 1000\,\xi && \text{(control value } - \text{ incident cost } - \text{ fuel cost)}\\
\text{s.t. } & \textstyle\sum_{w\in\mathcal{W}_g} f_{g,w} = |g| \quad \forall g && \text{(every free ship gets one destination)}\\
& \textstyle\sum_g\sum_{w\neq u(g)} f_{g,w} \le F_i - (\text{forced moves}) && \text{(movement budget)}\\
& \textstyle\sum_{(n,c,p)} b_{v,(n,c,p)} = 1 \quad \forall v && \text{(one option per zone)}\\
& \textstyle\sum_{(n,c,p)} n\, b_{v,(n,c,p)} = n_v, \quad \sum_{(n,c,p)} c\, b_{v,(n,c,p)} = c_v \quad \forall v && \text{(the option matches the moves)}\\
& \textstyle\sum_{v\in\mathcal{C}}\sum_{(n,c,p)} \rho_v(n,c,p)\, b_{v,(n,c,p)} \le \varepsilon_i\,(1-\ell^t/L) + \xi && \text{(escalation-risk budget)}\\
& f \text{ integer},\ b \text{ binary},\ \xi \ge 0.
\end{aligned}$$

Compared with §3.7, y becomes f, and the index n (a count for each type) becomes (n,c) (a navy and a coast guard strength level). Everything else is unchanged. In the code, the options of one posture (and, for assert, one coast guard level) are entered as a piecewise-linear function of the level with a breakpoint at every level that can occur, which is the same program with fewer variables.

Check: at every solve we checked, the program's objective equals the value of its plan recomputed by the game engine (`plan_value`), and it is never below the value of the greedy plan for the same state.

Run time and the game-ship scale: at full size, even an earlier version of this program that approximated the navy share took 13 seconds to 4 minutes per solve, and a one-year game needs 360 solves (two players, 180 periods). With each game ship standing for three real ships (§3.1), a year of MIP-BR against MIP-BR takes 30 to 39 minutes on one core. The experiments (about 12,400 games) therefore run as independent jobs on MIT's Engaging cluster.

3.9 The template game and the policies we compare

Template game: the full game has far too many strategies to solve for an equilibrium directly, so we also study a small strategic-form version. Each player picks one of 14 *templates*, which pair one of seven ways to split ~~70%~~two thirds (the share out of maintenance) of its fleet across North/Center/South (60/20/20, 40/40/20 or 34/33/33 percent and their permutations) with one posture (P or A) in all its zones. Ships are spread evenly within a region, Red's PLAN and CCG in proportion, and ~~fractional hulls are allowed~~only whole ships are used. With N_k^{dep} the deployable ships of type k rounded to a whole number and π_r the template's share for region r, the region counts are the largest-remainder rounding of $N_k^{\text{dep}}\pi_r$. Each region gets $\lfloor N_k^{\text{dep}}\pi_r \rfloor$ ships, and the ships left over go one at a time to the regions with the largest fractional parts. The same rule splits a region's ships over its zones. If both kept their templates, control would settle at $c_v = \phi_v$ (set $c_v^{t+1} = c_v^t$ in (1)), so a template pair's payoff is that settled control minus the incident cost, in calm seas at $\ell = 1$: $M_i(a_{\mathsf{B}}, a_{\mathsf{R}}) = \sigma_i \sum_{v\in\mathcal{C}} V_v^i \phi_v - \Lambda \sum_{v\in\mathcal{C}} \rho_v$. This gives a finite game $\Gamma = \langle \mathcal{I}, (S_i), (M_i) \rangle$ with $|S_i| = 14$, which we analyze with iterated elimination of strictly dominated strategies (IESDS), pure-NE enumeration, and the maximin and correlated-equilibrium LPs.

Policies (§4): *Random* picks moves and postures at random, as a benchmark. *Rule-based* (moderate commander) sends ships where value per unit of opponent strength is highest, asserts only where it believes it outnumbers the opponent and $\ell \le 2$, then pulls back until its expected incidents fit $\varepsilon = 0.15(1-\ell/L)$. *Greedy* is a fast heuristic for the MIP (same objective and budget, no solver). *MIP-BR*(ε) solves the MIP exactly. *Equilibrium* plays its part of the pure NE of Γ with the higher total payoff (our choice), steering free ships toward the template's split (more ships to a region's more valuable zones). *Maximin* draws a template every three periods from its maximin mixed strategy of Γ. The two template policies switch to patrol when $\ell > 3$ and drop assertions over their risk budget.

3.10 Resilient redesigns (Blue)

A redesign changes Blue's system. D1 is structural (an alternative path), D2 and D3 operational (monitoring, rapid response). Each is compared with the baseline D0 in every scenario. **D1 dispersed basing:** a third Blue port at Subic Bay, linked to SCB, LUZ and STS, giving alternative paths and shorter trips South. Blue starts 4/3/3 at YOK/GUA/Subic. **D2 redundant sensing:** Blue detects in every zone with probability at least 0.6, even in bad weather. **D3 de-escalation measures:** once $\ell \ge 3$, a quiet period lowers ℓ with probability 0.5 instead of 0.25, and assert hazards are halved, which helps Red as much as Blue. Costs ζ_D are in Table 1.

Simplifying assumptions: there are two unitary players, control is a force-ratio index, incidents are the only path to war, beliefs are last-seen counts, and best responses look one period ahead. Oilers follow a fixed rule and cannot be targeted, maintenance takes place only at home ports, and engineering casualties are independent from ship to ship.

4 Preliminary results

Note: every number in this section is new (one-year game, Workshop 2 fleets at one game ship per three real ships, readiness, exact MIP, $s_h = 0.12$). Workshop 1 values are shown in gray for comparison. Payoffs are discounted sums over 180 periods instead of 60, so they are larger in size than in Workshop 1.

§4.1 is analytical. The rest use 200 seeds per configuration (12,400 games in all). Unless stated, Red plays MIP-BR with $\varepsilon_R = 0.15$ in nominal weather.

4.1 Template game and a small example: patrol or assert, and sustainable restraint

IESDS reduces Γ from 14×14 to 3×2 (2×4). Blue keeps the center/south split (20/40/40) with assert and the even split (34/33/33) with either posture. Red keeps only the center/south split, with either posture. There are two pure NE. The *equilibrium* policy plays (Blue even split patrol, Red center/south assert), $(-22.96, 16.60)$. The other is (Blue center/south assert, Red center/south patrol), $(-11.52, 1.99)$. The welfare-maximizing correlated equilibrium puts all its weight on the first.

Fixing these splits (Blue even, Red center/south), each side only chooses P or A. Normalized so (P,P) $= (0,0)$, the 2×2 sub-matrix of Γ has (P,A) $= (-10.36, 2.19)$, (A,P) $= (1.06, -11.22)$ and (A,A) $= (-11.33, -10.49)$ (Blue, Red). For Red, asserting pays whatever Blue does ($2.19 > 0$ and $-10.49 > -11.22$), so assert is a dominant strategy. Blue gains a little by asserting against a patroller ($G_B = 1.06$), but against an asserting Red it loses less by patrolling (-10.36 against -11.33). The unique NE is (P,A): Red asserts and Blue patrols. In Workshop 1 this game had two pure NE, (A,P) and (P,A), and a mixed NE. The change comes from the fleets. Red now has about 1.6 times Blue's total strength, so it keeps control of a zone even when both sides assert.

This also ends the Workshop 1 restraint argument. There, falling back to the one-shot equilibrium after any assertion was a threat that made mutual patrol a subgame perfect equilibrium. Now Red earns 2.19 in the one-shot NE and 0 under mutual patrol, so that threat costs Red nothing, and mutual patrol cannot be held this way.

4.2 RQ1: control versus war risk, and the risk-budget game

B\R	cons. 0.05	mod. 0.15	aggr. 0.40
cons. 0.05	-378, 496	-1022, 1143	-1037, 1157
	war 0.01	0.01	0.01
mod. 0.15	579, -477	-430, 385	-748, 728
	war 0.01	0.26	0.17
aggr. 0.40	615, -506	20, -50	**-475, 86**
	war 0.00	0.22	**0.94**

Table 4: The risk-budget game: (u_B, u_R) and war rate when both sides choose ε (s.e. $\approx$ 6–13). Bold: the unique NE (-151, 39, war 0.34).

ε_B	control share	war rate	u_B
0.02	-0.63 (-0.25)	0.01 (0.07)	-1256
0.05	-0.50 (-0.18)	0.01 (0.11)	-1022
0.10	-0.38 (-0.10)	0.12 (0.14)	-795
0.15	-0.15 (-0.05)	0.26 (0.20)	-430
0.25	$+0.07$ (0.00)	0.25 (0.17)	-24
0.40	$+0.09$ ($+0.01$)	0.22 (0.25)	20

Table 5: The frontier: Blue MIP-BR with budget ε_B against Red MIP-BR ($\varepsilon_R = 0.15$). s.e. of control share ≤ 0.004, of war rate ≤ 0.03.

Frontier (Table 5)**:** the risk budget now buys far more control. Raising ε_B from 0.02 to 0.15 lifts Blue's control share from -0.63 to -0.15, while the war rate rises from 0.01 to 0.26. At 0.25 Blue reaches $+0.07$, and the war rate does not rise further. Past 0.25 the curve is flat. A tight budget costs Blue much more than in Workshop 1, because Red has about 1.6 times Blue's strength and holds every zone Blue leaves.

Risk-budget game (Table 4)**:** the structure is the same as in Workshop 1. Whatever the other side picks, "aggressive" pays each side the most, so (aggressive, aggressive) is the unique pure NE. Both sides would do better if both chose moderate ($-430, 385$ against $-475, 86$). This is still a prisoner's dilemma, but a much sharper one. Mutual aggression now ends in war in 94% of one-year games (34% of four-month games). The highest total payoff is again (Blue conservative, Red moderate).

4.3 RQ2: strategic versus heuristic play

Blue \ Red	rule-based (moderate)	MIP-BR	equilibrium
random	-816 / -0.36 / 0.21	-884 / -0.39 / 0.45	-819 / -0.37 / 0.34
rule-based (moderate)	-472 / -0.19 / 0.04	-713 / -0.32 / 0.12	-573 / -0.24 / 0.10
greedy	-183 / -0.03 / 0.13	-501 / -0.20 / 0.16	-287 / -0.09 / 0.10
MIP-BR	**-105** / $+0.02$ / 0.17	**-430** / -0.15 / 0.26	**-233** / -0.06 / 0.15
equilibrium	-568 / -0.25 / 0.06	-718 / -0.33 / 0.08	-588 / -0.26 / 0.10
maximin	-547 / -0.24 / 0.04	-710 / -0.33 / 0.08	-574 / -0.25 / 0.09

Table 6: Blue payoff u_B / control share / war rate (s.e. of u_B is 4–11). Nominal weather, design D0. The MIP-BR column is each policy's worst case. Bold: best Blue policy in each column.

MIP-BR is now Blue's best policy against every Red policy (Table 6), and it has the best worst case (-430). Greedy is second (-501). The gap between them is 71, about seven standard errors (a gap of 7, too small to separate). Greedy finds the MIP's plan in only 24% of 4,537 Blue decisions (67% of 1,524), and it breaks the risk budget in 11% (14%). The **equilibrium** and **maximin** plans fall far behind (-718 and -710 against MIP-BR) (equilibrium had the best worst case, -116). Both play one fixed split of the deployable two thirds of the fleet for the whole game. Unlike MIP-BR, they do not adapt to where Red is or to which of their own ships are available, and they end with a control share of -0.33 against -0.15 for MIP-BR. They do keep the war rate lowest (0.08 against MIP-BR). MIP-BR against MIP-BR has the highest war rate among the strong policies (0.26).

4.4 RQ3: robustness to weather and resilient designs

Design	calm	nom.	typh.	extr.	exp.	worst
D0 baseline	−458/.34	−430/.26	−399/.21	−366/.16	−423	−458
D1 dispersed	−474/.34	−421/.23	−394/.19	−368/.11	−422	−474
D2 sensing	−479/.34	−447/.28	−424/.28	−386/.12	−443	−479
D3 de-escalation	**−444**/.10	**−395**/.11	**−382**/.06	**−350**/.04	**−400**	**−444**

Table 7: Blue designs by weather scenario (MIP-BR vs. MIP-BR): u_B / war rate (s.e. ≈ 6–9).

We score each policy or design x by its expected payoff $\sum_\omega \pi_\omega J(x,\omega)$ and worst case $\min_\omega J(x,\omega)$.

Weather: as in Workshop 1, bad weather helps Blue. Every policy except MIP-BR with $\varepsilon = 0.40$ does worst in *calm* seas, since rough seas halve small-hull strength, which is about half of Red's strength but less than a quarter of Blue's, and storms pause operations and incidents (MIP-BR war rate 0.34 calm, 0.16 extreme). The top of the ranking is the same in all four scenarios: MIP-BR with $\varepsilon = 0.40$, then $\varepsilon = 0.15$, then greedy. Only the three weakest policies (rule-based, equilibrium, maximin) change places. The best policy on average ($\varepsilon = 0.40$, expected 3) is also best in its worst case (−88, extreme seas).

Designs: de-escalation measures (D3) are again best in expected and worst-case terms. They cut the war rate from 0.26 to 0.11 (nominal) (0.20 → 0.05) and raise Blue's payoff by 35. Red's payoff rises too (385 → 417), since fewer wars help both sides. We still charge them nothing, so part of this win is by assumption. Dispersed basing (D1) leaves Blue's payoff where it was (−421 against −430) and no longer cuts Red's (101 → 16), now 385 → 371. Redundant sensing (D2) is worse than the baseline (−447), paying its cost for no measurable benefit. **Limitations:** parameters are assumptions, policies look one period ahead, and each game ship stands for three real ships. Payoff differences below about 20 are not resolved with 200 games (s.e. 6–13).

5 Prototype

Current state: `wargame_gui.py` runs the model of §3 as a **GUI** (pick each side's plan, Blue's design and the weather, then step through turns or run many games), as **batch** runs, or as a **trace**. `experiments.py all` regenerates every number in §4.

How the code reflects the model: `Game.step` is the timeline of §3 with (1)–(4), `MIPBRPolicy` solves the MIP of §3.7 each period, and `MetaGame` builds Γ and runs IESDS, NE enumeration and both LPs. **Still to build:** learning agents, an attacker that reacts to Blue, Bayesian beliefs and multi-period policies (§6).

6 Plan for Workshop 2

Model: Bayesian beliefs, multi-period best response, sensitivity analysis on V, h, W, μ, and calendar-based weather from real data. Reformulate the movement budget F_i with considerations for oilers. **Learning:** a Red agent that learns from repeated play, and other learning methods on Γ. Do they converge, drift toward restraint, and exploit rule-based Blue? **Attacks and defenses:** three Red attacks are coded. A coast guard blockade at SCB and STS blocks Blue transits, jamming halves Blue's detection in one region, and false contacts appear in Blue's picture. We will add a Red attacker that reacts to Blue's posture and test randomized patrols, D1 and D2 as defenses. **Designs:** D0–D3 and combinations under attack and a long typhoon, and a cost for D3.

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